

Server Docker Setup Report

Summary

- Success/Failure Summary: Partial Success. Native Windows Docker Engine (v29.3.0) and Docker Compose (v5.1.0) installed and working perfectly for Windows containers. However, Linux containers are currently failing because WSL2 crashes with `E\_UNEXPECTED`, meaning the Linux backend cannot boot.
- Final Server Status: Ready for Windows containers. Needs a reboot to try unblocking WSL2 for Linux containers.
- Next Command for App Deployment (after unblocking Linux containers):
`cd C:\inetpub\pickpic; docker-compose up -d`

What Was Installed

- `Containers` Windows feature enabled
- Native Docker Engine CE (Community Edition) `v29.3.0` for Windows
- Docker Compose `v5.1.0` binary placed in `\$Env:ProgramFiles\docker\`
- Deployment directory created at `C:\inetpub\pickpic`

Key Commands Run

- `Get-WindowsFeature -Name Hyper-V, Containers`
- `wsl --status` and `wsl --update`
- `Invoke-WebRequest ... install-docker-ce.ps1`
- `docker run --rm mcr.microsoft.com/windows/nanoserver:ltsc2022 cmd.exe /c echo "Hello from Windows Container!"`
- `docker run --rm ubuntu:latest echo ...`

Status & Verifications

- Docker Service Status: `Running` (Docker Engine)
- Is Docker Working?: Yes. Successfully pulled and ran `nanoserver:ltsc2022`.
- Is Docker Compose Working?: Yes. `docker-compose version` returns `v5.1.0`.
- Linux Container Support: Not currently usable.
 - Reason: The native Windows `dockerd` requires WSL2 or a Hyper-V Linux VM to run Linux containers. The installed WSL2 Ubuntu instance on this server throws a Catastrophic Failure (`Wsl/Service/E\_UNEXPECTED`). Because of this deep-level virtualization failure, no Linux container system (Docker Desktop or otherwise) can start.

Remaining Blockers

- WSL2 Virtualization Failure: We enabled the `VirtualMachinePlatform` and `Containers` features, but WSL2 is critically failing when starting. As this is a VPS, nested virtualization issues or simply pending reboots are the likely culprits.

Recommended Next Steps

1. Reboot the Server: Mandatory step. A full restart is required to solidify the hypervisor and Virtual Machine Platform features we enabled.
2. Re-Test WSL: After rebooting, run `wsl -d Ubuntu` to ensure WSL2 works.

3. Switch to Docker Desktop (if WSL2 is fixed): Since native Docker only gives Windows containers, if you want easiest Linux container deployment on this server, install Docker Desktop once WSL2 is alive, or run a Linux VM.